

DECISION GOVERNANCE INFRASTRUCTURE

UNITED STATES PATENT AND TRADEMARK OFFICE

PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION:

BIDIRECTIONAL TEMPORAL ADMISSIBILITY SYSTEM FOR AUTOMATED
GOVERNANCE DECISIONS WITH RETROSPECTIVE TRAJECTORY
COHERENCE SCORING, PROSPECTIVE REGIME TRANSITION RISK
EVALUATION USING HIDDEN MARKOV MODEL TRANSITION MATRICES,
AND DUAL-SOURCE UNBIASED HISTORY ANALYSIS

Docket Number:	OMNIX-PAT-2026-014
Inventor:	Harold Alberto Nunes Rodelo
Applicant:	OMNIX QUANTUM LTD (United Kingdom)
Entity Status:	Micro Entity — 37 C.F.R. § 1.29
Filing Basis:	35 U.S.C. § 111(b) — Provisional Application
Date Prepared:	April 19, 2026
Date of Filing:	April 19, 2026
Related Applications:	OMNIX-PAT-2026-001 (Governance Architecture), OMNIX-PAT-2026-012 (TIE), OMNIX-PAT-2026-

This document constitutes a Provisional Patent Application filed pursuant to 35 U.S.C. § 111(b). It establishes a priority date and grants twelve (12) months from the filing date to submit a corresponding non-provisional application. CONFIDENTIAL — NOT FOR DISTRIBUTION.

FIELD OF THE INVENTION

The present invention relates to automated decision governance systems, and more particularly to a bidirectional temporal admissibility system that determines whether a proposed governance decision is temporally admissible by simultaneously evaluating **retrospective trajectory coherence** — whether the proposed decision is consistent with the recent history of the system — and **prospective trajectory implication** — whether the proposed decision is consistent with the likely near-future trajectory of the system as estimated by a Hidden Markov Model regime transition matrix. The combined bilateral temporal assessment constitutes a governance checkpoint that extends admissibility determination from the current instant to both the recent past and the probable near future.

BACKGROUND OF THE INVENTION

Multi-checkpoint governance pipelines for automated decision systems evaluate each proposed decision against a set of criteria reflecting the current state of the system: signal quality, probabilistic confidence, risk exposure, compliance thresholds, and coherence metrics. This evaluation is instantaneous — it considers the system's current state but not its trajectory through time.

The temporal admissibility gap has two dimensions:

Dimension A — Retrospective: A system may generate a nominally valid decision that contradicts the direction, regime, and signal pattern established by its recent history. A BUY decision generated after 15 consecutive cycles of bearish signals and declining regime quality is nominally valid if it passes all current-state checkpoints. It is temporally inadmissible because it is incoherent with the trajectory from which it emerged.

Dimension B — Prospective: A system may generate a nominally valid decision that is likely to become inadmissible within a small number of future cycles because the detected regime is likely to transition to an adverse regime. Approving a BUY decision in a regime that has an 80% probability of transitioning to BEARISH within 5 cycles approves an action that will, with high probability, be contradicted by the system's own future signals.

Existing governance systems address neither dimension. No prior art checkpoint pipeline performs retrospective trajectory coherence analysis or prospective HMM-based regime transition risk evaluation as a condition of decision admissibility.

SUMMARY OF THE INVENTION

The present invention provides a Bidirectional Temporal Admissibility System comprising two integrated components that together enforce temporal coherence of automated governance decisions:

TCV	Trajectory Coherence Validator	A retrospective module evaluating proposed decisions against three weighted dimensions of recent behavioral trajectory: direction coherence (40%), regime-action alignment (35%), and signal stability (25%). Uses dual-source history analysis combining shadow_trade_events (blocked decisions) and paper_trading_trades (approved decisions) to eliminate survivorship bias from the historical record.
FTI	Forward Trajectory Implication	A prospective module evaluating proposed decisions against two forward-looking risk dimensions: (1) HMM Regime Transition Risk — probability that the current regime transitions to an adverse regime within a configurable horizon; (2) Trajectory Terminus Risk — probability that continuing current trajectory leads to a governance-prohibited state within a configurable horizon.

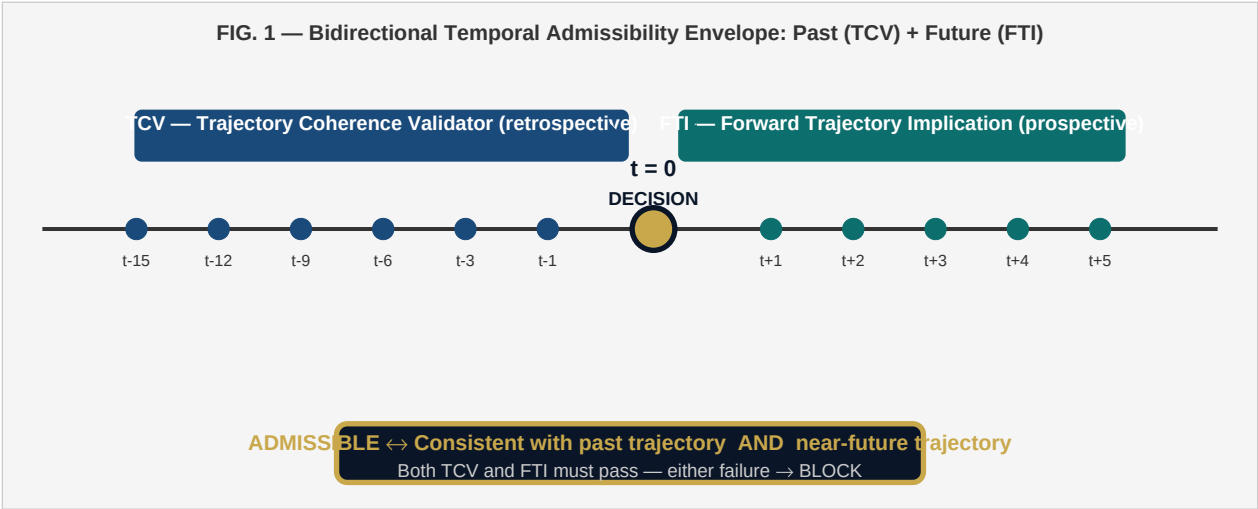


FIG. 1 — Bidirectional Temporal Envelope: TCV evaluates past trajectory; FTI projects future trajectory

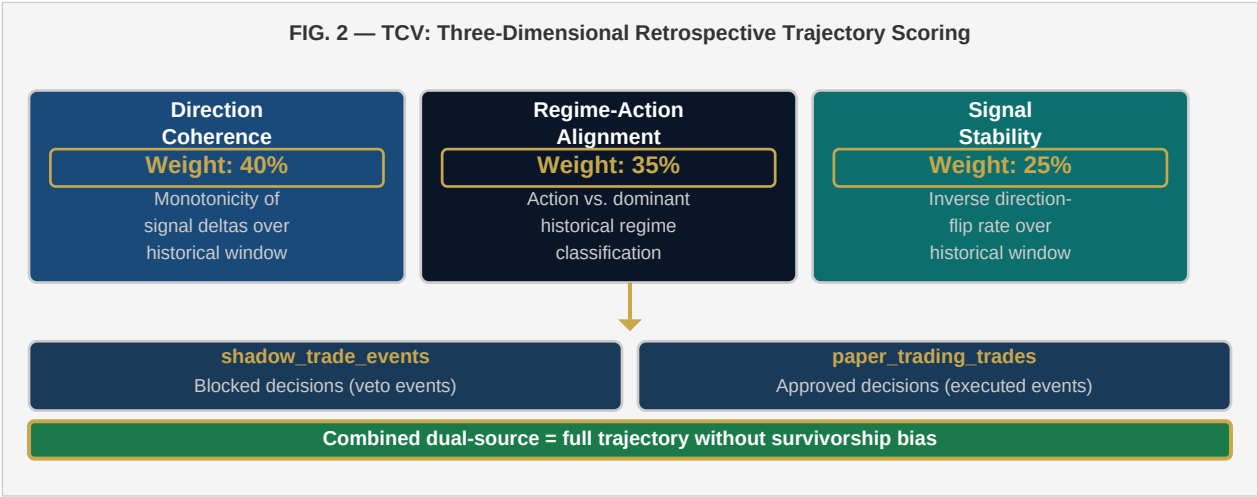


FIG. 2 — TCV Three-Dimensional Scoring with Dual-Source Unbiased History Analysis

DETAILED DESCRIPTION OF THE INVENTION

I. Trajectory Coherence Validator (TCV)

The TCV evaluates a proposed decision against the recent behavioral trajectory of the governance system across three independently weighted dimensions:

Dimension	Weight	Measurement	BLOCK Condition
Direction Coherence	40%	Monotonicity of signal deltas over history	Proposed direction contradicts majority of recent signal deltas
Regime-Action Alignment	35%	Consistency of proposed action with decision historical regime classification (e.g., BUY in	action inconsistent with regime classification (e.g., BUY in persistent BEAR
Signal Stability	25%	Inverse of direction-flip rate over history	High flip rate indicates unstable signal environment; action blocked

Dual-Source History Analysis: The TCV eliminates survivorship bias from historical trajectory analysis by combining two data sources: shadow_trade_events (records of all BLOCKED decisions, including the proposed action that was rejected and the reason for rejection) and paper_trading_trades (records of all APPROVED decisions that proceeded to execution). The combination provides a complete, unbiased trajectory that includes both executed and counterfactual decisions.

II. Forward Trajectory Implication (FTI)

The FTI evaluates the near-future implications of a proposed decision across two prospective risk dimensions:

HMM Regime Transition Risk: Using the Hidden Markov Model transition probability matrix calibrated to the governance system's historical regime sequence, the FTI computes the probability that the current regime transitions to an adverse regime (BEARISH, DOWNTREND, VOLATILE, STRONG_SELL) within a configurable forward horizon (default: 5 cycles). A BUY decision proposed in a regime with high probability of adverse transition is prospectively inadmissible regardless of current-state checkpoint results.

Trajectory Terminus Risk: The FTI projects the current behavioral trajectory forward and evaluates the probability that continuing on this trajectory would produce a governance-prohibited state (e.g., maximum drawdown breach, regulatory threshold violation) within the forward horizon. A proposed decision that accelerates convergence toward a prohibited terminus state is prospectively inadmissible.

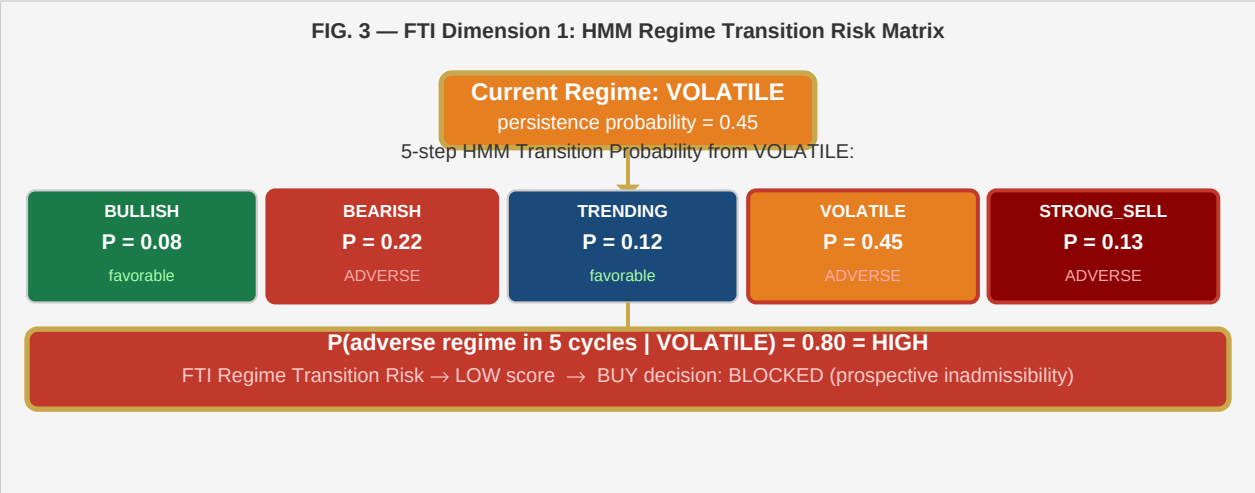


FIG. 3 — HMM Regime Transition Risk: VOLATILE regime → 80% probability adverse transition in 5 cycles

III. Bilateral Admissibility Determination

The bilateral admissibility determination combines TCV and FTI results using a conjunction logic: a proposed decision is temporally admissible only if both TCV returns ADMISSIBLE and FTI returns PASSED. Either

module independently failing is sufficient to block the proposed decision. Both modules incorporate fail-safe (PASS-THROUGH) behavior: if a module cannot complete evaluation due to data unavailability, evaluation proceeds but the fail-safe event is logged for audit and threshold recalibration.

FIG. 4 — Bilateral Admissibility Logic: TCV × FTI → Governance Outcome		
TCV Result	FTI Result	Bilateral Decision
ADMISSIBLE	PASSED	✓ PROCEEDS — bilateral admissibility confirmed
NOT ADMISSIBLE	PASSED	✗ BLOCKED — retrospective trajectory violation
ADMISSIBLE	NOT PASSED	✗ BLOCKED — prospective regime transition risk
NOT ADMISSIBLE	NOT PASSED	✗ BLOCKED — bilateral compound failure
PASS-THROUGH	PASSED	■ PROCEEDS — TCV fail-safe mode (logged)
ADMISSIBLE	PASS-THROUGH	■ PROCEEDS — FTI fail-safe mode (logged)

FIG. 4 — Bilateral Admissibility Logic: Six outcome combinations; only TCV-ADMISSIBLE + FTI-PASSED allows execution

CLAIMS

1. A bidirectional temporal admissibility system for automated decision governance comprising: a Trajectory Coherence Validator (TCV) configured to evaluate a proposed decision against the retrospective behavioral trajectory of the governance system across at least three weighted dimensions including direction coherence, regime-action alignment, and signal stability; a Forward Trajectory Implication (FTI) module configured to evaluate the prospective implications of the proposed decision using a Hidden Markov Model regime transition matrix to estimate the probability that the current regime transitions to an adverse regime within a configurable forward horizon; and a bilateral admissibility determination unit configured to approve the proposed decision only when both the TCV returns an ADMISSIBLE result and the FTI returns a PASSED result, and to issue a BLOCK verdict when either module returns an adverse result.
2. The system of claim 1, wherein the Trajectory Coherence Validator evaluates direction coherence by computing the monotonicity of signal deltas over a configurable historical window; evaluates regime-action alignment by determining the consistency of the proposed action with the dominant regime classification observed over the historical window; and evaluates signal stability by computing the inverse of the direction-flip rate over the historical window.
3. The system of claim 1, wherein the Trajectory Coherence Validator is configured to perform dual-source history analysis by combining: a first data source comprising records of blocked governance decisions including the proposed action and the reason for rejection; and a second data source comprising records of approved governance decisions that proceeded to execution; wherein the combination eliminates survivorship bias from the historical trajectory record.
4. The system of claim 1, wherein the Forward Trajectory Implication module computes a regime transition risk score by: identifying the current detected regime; retrieving from the Hidden Markov Model transition matrix the transition probabilities from the current regime to each identified adverse regime; computing the aggregate probability of adverse regime transition within the configurable forward horizon; and classifying the regime transition risk as LOW, MEDIUM, or HIGH based on the computed aggregate probability.
5. The system of claim 1, wherein the Forward Trajectory Implication module further comprises a trajectory terminus risk component configured to project the current behavioral trajectory forward and determine the probability that the projected trajectory reaches a governance-prohibited state within the configurable forward horizon.
6. The system of claim 1, wherein each of the Trajectory Coherence Validator and Forward Trajectory Implication module is configured to enter a pass-through mode when unable to complete its evaluation due to data unavailability; wherein pass-through mode permits the proposed decision to proceed subject to logging the fail-safe event; and wherein repeated fail-safe events trigger threshold recalibration review.
7. The system of claim 1, wherein the bidirectional temporal admissibility system constitutes a governance checkpoint in a multi-checkpoint governance pipeline, such that a BLOCK verdict from the bilateral admissibility determination unit is independently sufficient to prevent the proposed decision from reaching an execution environment, regardless of the pass/block verdicts of any other checkpoint in the pipeline.
8. A method for determining temporal admissibility of proposed automated governance decisions comprising: evaluating a proposed decision against the retrospective behavioral trajectory of the governance system to produce a trajectory coherence result; evaluating the proposed decision against the prospective regime transition probabilities derived from a Hidden Markov Model transition matrix to produce a forward trajectory result; issuing an admissibility determination that approves the proposed decision only when both the trajectory coherence result is ADMISSIBLE and the forward trajectory result is PASSED; and blocking the proposed decision when either evaluation result is adverse.

9. The method of claim 8, wherein evaluating the retrospective behavioral trajectory comprises combining records of blocked decisions and records of approved decisions into a unified trajectory history that includes both executed and counterfactual governance decisions.

10. The method of claim 8, wherein the Hidden Markov Model transition matrix is calibrated to the historical regime sequence of the governance system and updated on a configurable refresh schedule, and wherein adverse regimes are configurable per deployment domain to accommodate domain-specific governance requirements in financial trading, insurance underwriting, medical AI, real estate governance, and energy systems.

ABSTRACT

A Bidirectional Temporal Admissibility System for automated decision governance extends admissibility determination from the current instant to both the recent past and the probable near future. A Trajectory Coherence Validator (TCV) evaluates proposed decisions against three weighted retrospective dimensions — direction coherence (40%), regime-action alignment (35%), and signal stability (25%) — using dual-source history analysis combining blocked and approved decision records to eliminate survivorship bias. A Forward Trajectory Implication (FTI) module evaluates prospective regime transition risk using a Hidden Markov Model transition matrix and trajectory terminus risk analysis. A bilateral admissibility determination approves a proposed decision only when both TCV and FTI concur; either module independently can block execution. The system provides a governance guarantee that a decision is coherent with the trajectory from which it emerged and not likely to become inadmissible within the near-future projection horizon — a temporal admissibility dimension absent from all prior art governance checkpoint pipelines. The system is domain-agnostic and applies to financial trading, insurance underwriting, medical AI, real estate, and energy governance.

Application Reference: OMNIX-PAT-2026-014 | Inventor: Harold Alberto Nunes Rodelo | OMNIX QUANTUM LTD, United Kingdom | ©
2026 OMNIX Quantum Ltd. CONFIDENTIAL.